

LUMEN ARCHITECT

A two-player asymmetric multiplayer network game

CSCI 22 - Introduction to Programming II

Final Project, Second Semester 2025-2026

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Description

Lumen Architect is a two-player asymmetric multiplayer network game played across two machines on the same local network (or two terminals on a single machine for testing). Players take on opposing but cooperating roles in a dark platform world threaded through with light, lore fragments, and corrupted hazards.

The Wanderer explores a side-scrolling platform world from inside the game using the keyboard. Their goal is to traverse three Acts of increasing difficulty, collect lore fragments that unlock new abilities (melee, projectile, dodge, wall-cling, shadow dash, and more), and finally enter the Boss arena to destroy all four Cores before the time runs out or their health drops to zero.

The Apprentice plays from a separate window using the mouse. They place and remove platforms to help the Wanderer traverse hazards, control the radius and intensity of the light source the Wanderer relies on to see, and during the Boss fight they pick from five different boss attack patterns to challenge the Wanderer.

Cooperation is forced by the asymmetric design: the Apprentice cannot win without the Wanderer reaching the cores, and the Wanderer cannot survive without the Apprentice's light and platforms.

Game Mechanics

Server-authoritative networking

The game runs at 60 frames per second with a server-authoritative network model. **GameServer** is the single source of truth for all shared state - Core health, victory conditions, fragment collection, boss attack cooldowns - and broadcasts state packets to both clients every 16 ms. Clients never trust their own local values for any state the server owns.

Light, darkness, and visibility

Most of the world is dark. The Apprentice's light source illuminates a radius of platforms and entities; outside that radius the Wanderer cannot see hazards or even some platforms. The Wanderer must learn to play with limited information and trust their partner's positioning of the light.

Acts and progression

The campaign has four levels:

- **Act 1** - tutorial; basic traversal; first lore fragments.
- **Act 2** - wall-cling traversal, breakable walls, altars.
- **Act 3** - shadow-dash and Iron-fragment puzzles.
- **Boss arena** - top-down combat against the four Cores.

Lore fragments and ability unlocks

Lore fragments are scattered across the Acts; collecting them unlocks new Wanderer abilities. The full unlock list is:

- **MELEE** - close-range J attack.
- **PROJECTILE** - charged K-key ranged shot.
- **DODGE** - L-key invincibility roll.
- **WALL_CLING** - cling to and climb vertical walls.
- **SHADOW_DASH** - SHIFT-key phase through short gaps.
- **EMBER, IRON** - boss-fight stat boosts.
- **RADIANT_COLLAPSE** - boss-only full-arena light reveal.
- **VEIL, ECHO, TETHER, SHADOW_STEP** - traversal upgrades.

How to Play

Compiling

Open a Command Prompt or Terminal in the project root folder and run:

```
javac -Xlint:none *.java
```

This compiles all .java source files in place. No build tool, IDE, or external library is required.

Step 1 - Start the server (always FIRST)

In the project root, run:

```
java GameServer
```

The server listens on port **9876** and waits for two clients to connect.

Step 2 - Start the first client (Wanderer)

In a second terminal (same machine OK for testing), run:

```
java GameStarter
```

When the client window appears, type the server's IP address in the "Enter server IP..." field and click **CONNECT**. Use `localhost` or `127.0.0.1` if the server is on the same machine. In the lobby, click the **WANDERER** card and click READY.

Step 3 - Start the second client (Apprentice)

In a third terminal (or on a second machine on the same LAN), run:

```
java GameStarter
```

Connect to the same server IP, then click the **APPRENTICE** card and click READY. Once both players are ready, the game starts automatically.

Step 4 - Quit

Each client window has a Quit button on the in-game pause menu (press ESC). The server window can be closed with Ctrl+C in its terminal.

Wanderer Controls (keyboard)

Movement

- A or LEFT** Move left
- D or RIGHT** Move right
- SPACE or W** Jump

Combat

- J** Melee attack (*requires MELEE fragment*)
- Hold K, release** Charge projectile (*requires PROJECTILE fragment*)
- L** Dodge roll (*requires DODGE fragment*)

Special abilities

- E** Light pulse / level-ready signal
- SHIFT** Shadow dash (*requires SHADOW_DASH fragment*)
- SHIFT (hold 2 s)** Radiant Collapse (*BOSS phase only - full-arena reveal*)

System

- ESC** Pause / unpause menu
- F** Open / close the Fragment Library overlay

Apprentice Controls (mouse)

- Mouse motion** Move the light source
- Left click** Place a block of the currently selected type
- Right click** Remove the block under the cursor
- Mouse wheel** Adjust the light radius (smaller / larger)
- TAB** Cycle through available block types
- E** Signal level-ready
- Q** Radius burst (BOSS phase)
- T** Toggle the light on / off

All placements consume the per-level Block Budget. Watch the budget counter on the HUD; it refills at the start of each Act. Available block types depend on the level: BRICK, SLIDE, SPRING, WALL, CRUMBLE, INVISIBLE, and MIMIC.

Tips and Strategy

For the Wanderer

- Communicate with your Apprentice. They cannot see what you see; describe your situation.
- Don't rush blind. Outside the light radius, hazards are invisible and deadly.
- Hold K to charge a projectile fully before releasing - the longer the charge, the more damage and range.
- Conserve your dodge cooldown for the boss arena; you'll need it.

For the Apprentice

- Block budget is finite. Place sparingly and remove blocks you no longer need.
- MIMIC blocks look like BRICK but collapse 800 ms after first contact; use them as traps when the Wanderer expects safety.
- Keep the light just ahead of the Wanderer, not centred on them - they need to see what's coming.
- During boss fights, vary your attack types; each has a per-type cooldown but together they stack pressure.

Sources Cited

CSCI 22 course materials, Second Semester 2025-2026:

- [1] Module 1a - Modifiers
- [2] Module 1b - Interfaces
- [3] Module 1c - Abstract Classes
- [4] Module 1d - Inner and Anonymous Classes
- [5] Module 2a - Event Handling
- [6] Module 2b - Debugging
- [7] Module 2c - UML Diagrams
- [8] Module 3a - Graphics in Java
- [9] Module 3b - More Graphics in Java
- [10] Module 3c - Collision
- [11] Module 4a - Threads
- [12] Module 4b - Key Bindings
- [13] Module 4c - Networking in Java

Companion graphics tutorials (YouTube):

- [14] "Java Graphics Programming Tutorial - Shapes, Paths, Curves, and Transformations."
<https://youtu.be/zCiMlbu1-aQ>
- [15] "Animation Introduction (Java Graphics)." <https://youtu.be/pdtEB3R4MZI>

External technical sources: See the project's COA and the in-source `// Citations` blocks for full citations of external sources used for networking optimisation, threading idioms, advanced graphics composition, collision response, and game architecture patterns that go beyond the course module scope.